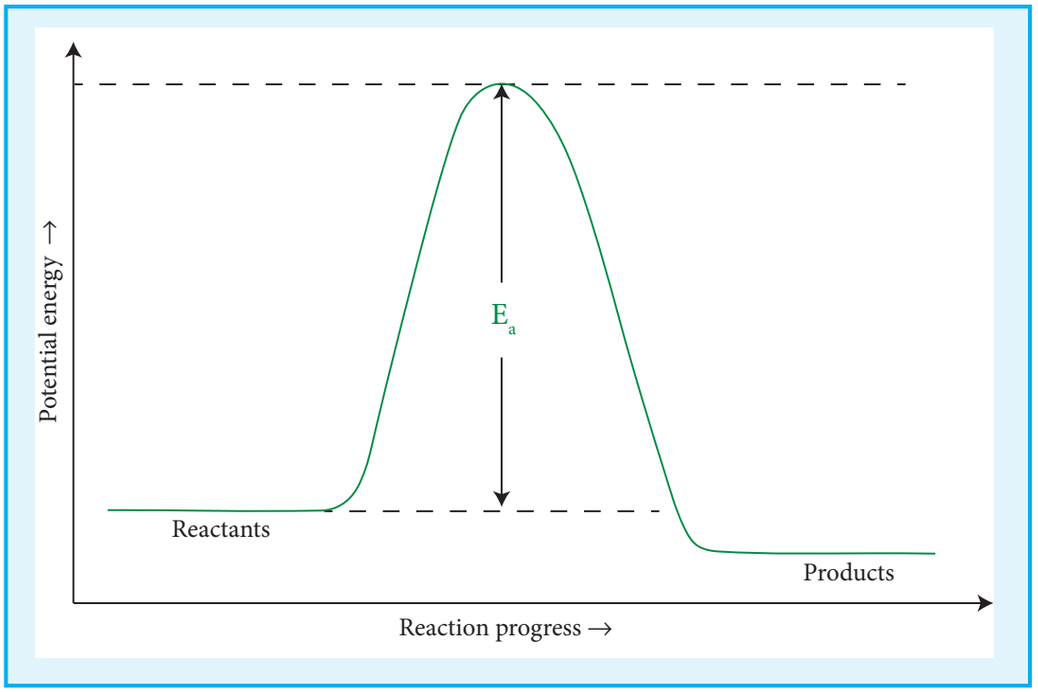


Fig 7.5 progress of the reaction

If we consider that, the reaction between A_2 and B_2 molecules proceeds through collisions between them, then the rate would be proportional to the number of collisions per second.

Rate $\propto$ number of molecules colliding per litre per second (collision rate)

The number of collisions is directly proportional to the concentration of both A_2 and B_2.

Collision rate $\propto$ $[A_2][B_2]$

Collision rate = $Z\,[A_2][B_2]$

Where, Z is a constant.

The collision rate in gases can be calculated from kinetic theory of gases. For a gas at room temperature (298K) and 1 atm pressure, each molecule undergoes approximately 10^9 collisions per second, i.e., 1 collision in 10^{-9} second. Thus, if every collision resulted in reaction, the reaction would be complete in 10^{-9} second. In actual practice this does not happen. It implies that all collisions are not effective to lead to the reaction. In order to react, the colliding molecules must possess a minimum energy called activation energy. The molecules that collide with less energy than activation energy will remain intact and no reaction occurs.

Fraction of effective collisions (f) is given by the following expression

$$f = e^{\frac{-E_a}{RT}}$$

To understand the magnitude of collision factor (f), Let us calculate the collision factor (f) for a reaction having activation energy of $100\ kJ\ mol^{-1}$ at 300K.

$$f = e^{-\left(\frac{100\times10^3\ J\ mol^{-1}}{8.314J\ K^{-1}\ mol^{-1}\ \times\ 300K}\right)}$$

$$f = e^{-40} \approx 4 \times 10^{-18}$$

Thus, out of 10^{18} collisions only four collisions are sufficiently energetic to convert reactants to products. This fraction of collisions is further reduced due to

XII U7 kinetics - Jerald Folder.indd 218 04-01-2021 13:33:07